



7583 - Unique H-alpha imaging of a $z=3.95$ starbursting protocluster

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
H-alpha NB imaging				
	1	NB H-alpha imaging	NIRCam Imaging	(1) LH-SBC3
Repeat observation				
	2	NB imaging repeat	NIRCam Imaging	(1) LH-SBC3

ABSTRACT

We propose to obtain highly sensitive H-alpha imaging of the starbursting protocluster LH-SBC3 at $z=3.95$ with the NIRCam F323N filter. This protocluster hosts an integrated SFR of 4000 $M_{\odot}/\text{yr}$ in its compact 100 kpc core, making it one of the most distant and intensely star-forming structures known to date. Fortunately, the H-alpha emission at $z=3.95$ falls perfectly within the narrowband (NB) filter F323N, making LH-SBC3 the only known protocluster where one can conduct NB H-alpha imaging at $z\sim 4$. This is a unique opportunity to spectroscopically confirm a large number of protocluster galaxies and investigate the early phases of structure formation. Our primary goal is to robustly characterize star formation activity, stellar mass assembly, and quantify the environmental effects in a $z\sim 4$ massive protocluster. Second, the supporting broadband imaging will pioneer the detection of intracluster light (ICL) at $z\sim 4$ and place a stringent constraint on the ICL fraction, providing crucial insights into the origin of ICL.

OBSERVING DESCRIPTION

We use NIRCam/F323N imaging to observe H-alpha emission of LH-SBC3 at $z=3.95$. At $z=3.95$, H-alpha is redshifted into the F323N filter with almost a perfect fit. In order to measure the continuum underlying H-alpha lines, we additionally request relatively short integration with the F322W2 filter. Thanks to the extra-wide band of F322W2, the observations will also enable us to detect the ICL or strongly constrain the ICL fraction in LH-SBC3. Alongside with the primary LW imaging, we request parallel SW imaging of F115W, F150W and F200W, which will bracket UV continuum and Balmer break of cluster galaxies. Additionally, we request F210M+F444W imaging to robustly constrain the SEDs for all sources, which is essential for discriminating interlopers, identifying quiescent members, and constraining dust attenuation. We adopt the NIRCam module "ALL", subarray "FULL", primary dither type "INTRAMODULEBOX 4" with readout pattern "MEDIUM 8". We adopt an offset to place the target in the deepest area of the mosaic. The dither type "INTRAMODULEBOX 4" will remove the gaps in SW images, meanwhile it ensures the full structure being observed with the deepest integration in both SW and LW.

Proposal 7583 - Targets - Unique H-alpha imaging of a $z=3.95$ starbursting protocluster

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	LH-SBC3	RA: 10 41 53.0137 (160.4708904d) Dec: +58 45 23.61 (58.75656d) Equinox: J2000		
	Comments: The target is in the center of the protocluster LH-SBC3 at $z=3.95$. The virial radius of the cluster is 40" (210 kpc). Category=Clusters of Galaxies Description=[High-redshift clusters] Extended=YES				

Proposal 7583 - Observation 1 - Unique H-alpha imaging of a z=3.95 starbursting protocluster

Observation	Proposal 7583, Observation 1: NB H-alpha imaging								Wed May 13 13:00:12 GMT 2026		
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
	Comments: We adopt an offset to place the center of the cluster in the center of a short wavelength detector of module B, guaranteeing that the SW mosaic can fully cover the whole structure without leaving a gap.										
Diagnostics	(NB H-alpha imaging (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates		Targ. Coord. Corrections		Miscellaneous				
	(1)	LH-SBC3	RA: 10 41 53.0137 (160.4708904d)								
			Dec: +58 45 23.61 (58.75656d)								
			Equinox: J2000								
	Comments: The target is in the center of the protocluster LH-SBC3 at z=3.95. The virial radius of the cluster is 40" (210 kpc).										
	Category=Clusters of Galaxies										
	Description=[High-redshift clusters]										
	Extended=YES										
Template	Module		Subarray			Target Placement					
	ALL		FULL			Module gap (large extended source)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F323N+F322W2	MEDIUM8	10	1	4	4	4208.814	164422	
	2	F150W	F323N+F322W2	MEDIUM8	10	1	4	4	4208.814	164422	
	3	F200W	F323N+F322W2	MEDIUM8	10	1	4	4	4208.814	164422	
	4	F200W	F322W2	MEDIUM8	5	1	4	4	2061.46	164423	
	5	F210M	F444W	MEDIUM8	5	1	4	4	2061.46	176728	
Special Requirements	Offset 65.0 arcsec, 23.0 arcsec										

Proposal 7583 - Observation 2 - Unique H-alpha imaging of a z=3.95 starbursting protocluster

Observation	Proposal 7583, Observation 2: NB imaging repeat									Wed May 13 13:00:12 GMT 2026	
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(1)	LH-SBC3	RA: 10 41 53.0137 (160.4708904d)								
			Dec: +58 45 23.61 (58.75656d)								
			Equinox: J2000								
			Comments: The target is in the center of the protocluster LH-SBC3 at z=3.95. The virial radius of the cluster is 40" (210 kpc). Category=Clusters of Galaxies Description=[High-redshift clusters] Extended=YES								
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEBOX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F323N+F322W2	MEDIUM8	10	1	4	4	4208.814	164422	
	2	F150W	F323N+F322W2	MEDIUM8	10	1	4	4	4208.814	164422	
	3	F200W	F323N+F322W2	MEDIUM8	10	1	4	4	4208.814	164422	
	4	F200W	F322W2	MEDIUM8	5	1	4	4	2061.46	164423	
	5	F210M	F444W	MEDIUM8	5	1	4	4	2061.46	176728	
Special Requirements	Fiducial Point Override NRCB1_FULL										